BLUESTOCK FINTECH

Mutual Fund Analytics Capstone

Final Project Report

An end-to-end data pipeline, analytics, and dashboarding project covering
40 mutual fund schemes, ~64,000 NAV records, and ~33,000 investor transactions

Prepared by: Tejas Bhole
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1. Executive Summary

This report presents the findings of a comprehensive mutual fund analytics capstone project undertaken for Bluestock Fintech. The project builds an end-to-end pipeline — from raw data ingestion through cleaning, exploratory analysis, advanced risk analytics, and an interactive Power BI dashboard — covering 40 mutual fund schemes across 10 major Indian fund houses, approximately 64,000 daily NAV observations, and roughly 33,000 individual investor transactions spanning 2022–2026.

The analysis confirms several expected patterns in the Indian mutual fund industry while surfacing a few findings that merit closer attention. Equity funds substantially outperform Debt funds on a returns basis (15.5% vs 6.3% average 1-year return) but carry proportionally higher risk, and this risk-return relationship holds consistently across the fund universe. SBI Mutual Fund leads the industry in assets under management, and Small Cap funds dominate the top-performer rankings by 3-year return.

On the investor side, SIP inflows have grown steadily industry-wide, and the platform's own investor base is generating strong transaction volume — but a striking 97.8% of investors with 6+ SIP transactions show average contribution gaps exceeding 35 days, a finding that warrants product and retention investigation. Sector concentration analysis (HHI) also flags several equity funds as more concentrated than their category label might suggest.

This report is organized to walk through the full analytical journey: data architecture and cleaning methodology, exploratory findings, performance and risk analysis, dashboard highlights, and forward-looking recommendations, closing with an honest account of the project's current limitations.

2. Project Objectives & Deliverables

2.1 Objectives

- O1 — Ingest and consolidate raw mutual fund, transaction, NAV, and performance data into a structured database.
- O2 — Clean and validate data: amount thresholds, KYC status, date formats, duplicate removal.
- O3 — Build a relational (star-schema) data model suited for analytics.
- O4 — Conduct exploratory data analysis across funds, investors, and transaction patterns.
- O5 — Compute fund performance and risk metrics: Sharpe, Sortino, Alpha, Beta, VaR/CVaR, HHI.
- O6 — Analyze investor behavior: cohort analysis, SIP continuity, demographic segmentation.
- O7 — Build an interactive Power BI dashboard spanning industry, performance, investor, and SIP trend views.
- O8 — Deliver a fund recommendation engine based on investor risk appetite.

2.2 Deliverables

- D1 — Cleaned, structured SQLite database (bluestock_mf.db).
- D2 — Data cleaning & ETL scripts (scripts/).
- D3 — Exploratory Data Analysis notebook with 15 analyses and documented findings.
- D4 — Advanced Analytics notebook: VaR/CVaR, rolling Sharpe, cohort analysis, SIP continuity, HHI concentration.
- D5 — Fund recommender script (recommender.py).
- D6 — Power BI dashboard with 4 pages (Industry Overview, Fund Performance, Investor Analytics, SIP & Market Trends).
- D7 — Final report and presentation deck (this document and accompanying slides).

3. Data & ETL

The project consolidates five distinct raw data sources — fund master data, historical NAV, transaction logs, portfolio holdings, and industry-level AUM/SIP statistics — into a single SQLite database using a star-schema design: one date dimension, one fund dimension, and five fact tables.

3.1 Data Model

Table	Type	Rows	Purpose
dim_date	Dimension	1,608	Calendar attributes: year, month, quarter, weekday flag
dim_fund	Dimension	40	Fund master: scheme, category, expense ratio, risk category
fact_nav	Fact	64,320	Daily NAV per fund
fact_transactions	Fact	32,778	Investor-level SIP/Lumpsum/Redemption transactions
fact_performance	Fact	40	Returns, Sharpe, Sortino, Alpha, Beta, risk grade per fund
fact_portfolio	Fact	322	Stock-level holdings & sector weights per fund
fact_aum	Fact	90	Fund-house level AUM over time
fact_sip_industry	Fact	48	Industry-wide monthly SIP inflow statistics

3.2 Cleaning Methodology

Raw transaction data was validated and cleaned through a dedicated pipeline stage before loading:

- Amount validation — transactions with amount_inr ≤ 0 were dropped as invalid.
- KYC status validation — records with unexpected KYC status values outside the accepted set were flagged and removed, with counts logged.
- Date normalization — transaction_date was coerced to a consistent datetime format; unparseable dates were dropped.
- Duplicate removal — exact duplicate rows were identified and removed prior to load.

Each cleaning step logs the number of rows affected, giving a transparent before/after audit trail for every run of the pipeline.

3.3 Pipeline Architecture

The pipeline runs as a sequence of independent, single-responsibility scripts: data ingestion → per-domain cleaning (transactions, NAV history, performance) → database load → metrics computation. A master run_pipeline.py orchestrates the full sequence end-to-end, allowing the entire ETL and analytics process to be reproduced with a single command.

4. EDA Findings

Fifteen exploratory analyses were performed across NAV, AUM, SIP, fund performance, portfolio composition, and investor demographics. The most significant findings are summarized below, illustrated with selected charts.

4.1 Returns by Category

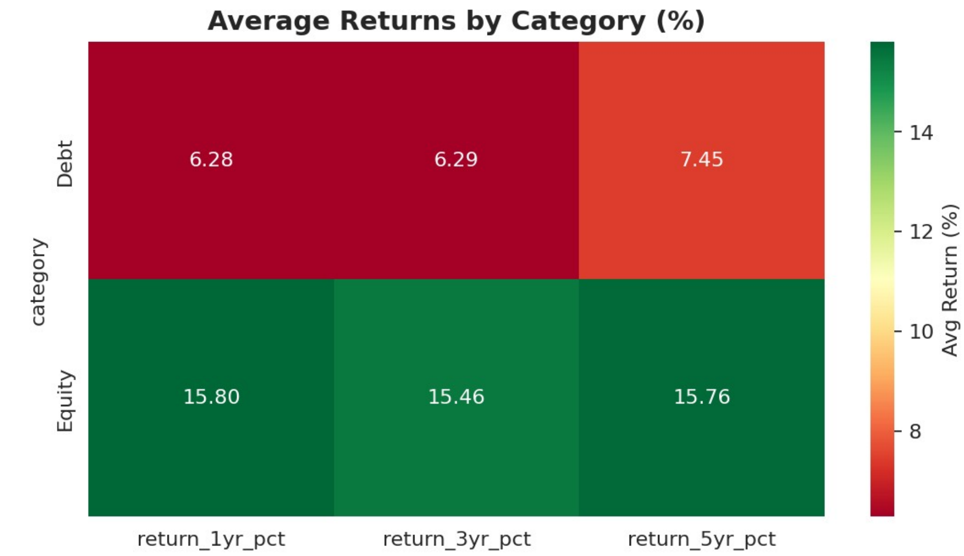


Figure 1 — Average 1/3/5-year returns by fund category

Equity funds average roughly 15.5% returns across the 1-, 3-, and 5-year horizons, more than double the 6-7% delivered by Debt funds. This gap is consistent and expected, reflecting the higher risk premium equity carries.

4.2 Risk vs Return Positioning

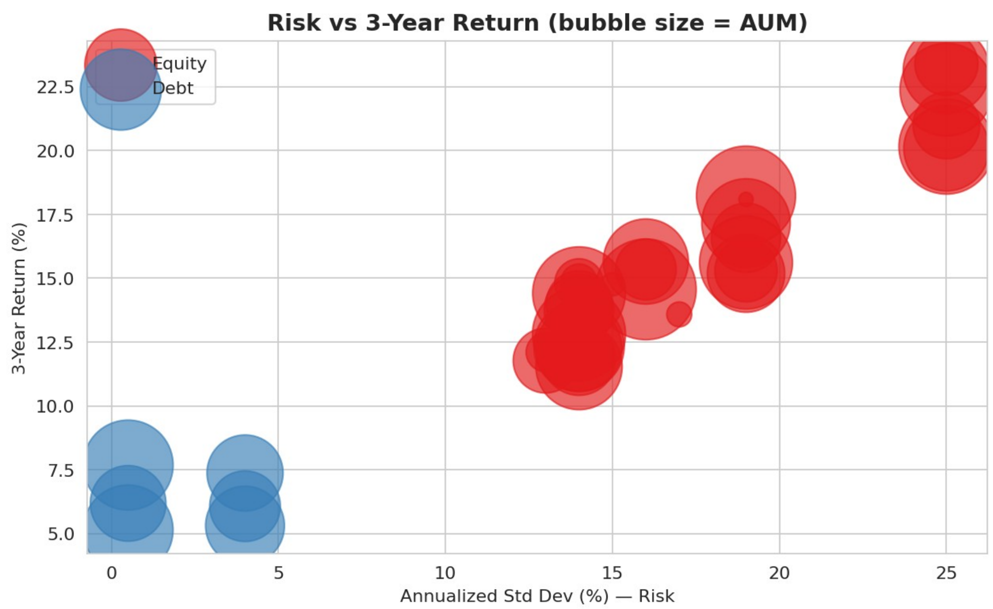


Figure 2 — Risk (annualized std. dev.) vs 3-year return, bubble size = AUM

The scatter shows a clear positive relationship between risk and return: funds with higher annualized volatility generally deliver higher 3-year returns, particularly within the Equity category. This is consistent with standard portfolio theory and gives confidence that the underlying data reflects realistic market dynamics.

4.3 SIP Inflow Trend

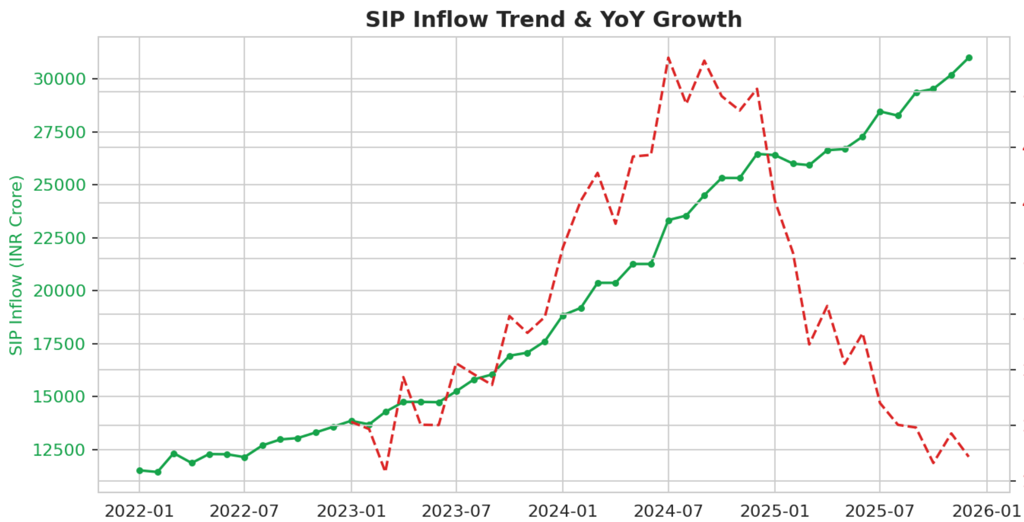


Figure 3 — Industry SIP inflow and year-over-year growth

Industry-wide SIP inflow rose from approximately ₹11,500 Cr to ₹31,000 Cr over the observed period. Year-over-year growth peaked around mid-2024 and has since moderated — a sign of a maturing market rather than slowing adoption, since absolute inflows continue to climb.

4.4 Top and Bottom Performing Funds

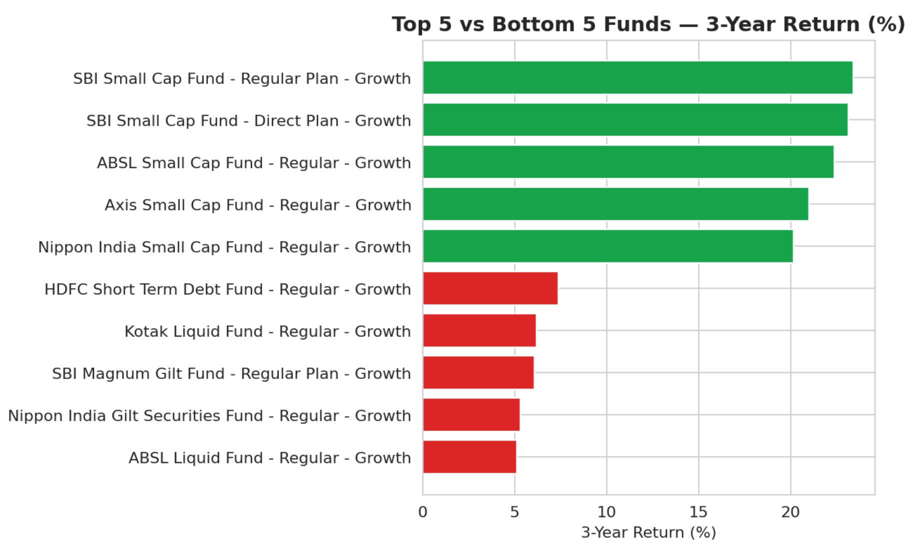


Figure 4 — Top 5 and bottom 5 funds by 3-year return

All five top-performing funds by 3-year return are Small Cap equity funds, while the bottom five are Debt, Gilt, and Liquid funds — a clean illustration of the risk-return tradeoff playing out at the individual fund level.

4.5 Correlation Structure

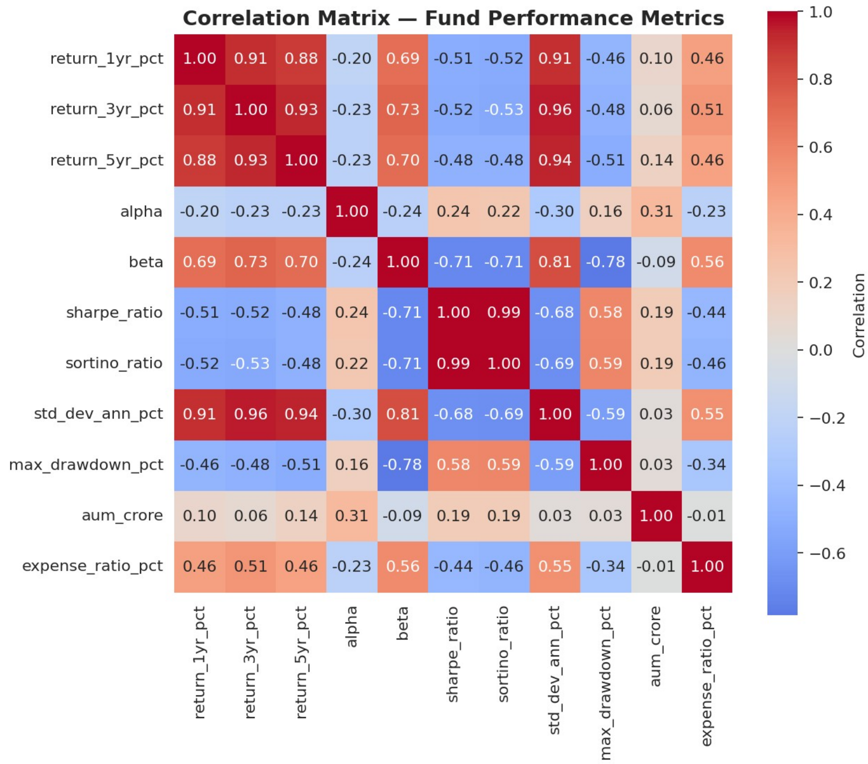


Figure 5 — Correlation matrix across fund performance metrics

Sharpe and Sortino ratios are nearly perfectly correlated (r ≈ 0.99) in this fund universe, indicating that downside risk closely tracks overall volatility here — either metric can be used interchangeably for risk-adjusted fund comparison without materially changing conclusions.

4.6 Portfolio Sector Allocation

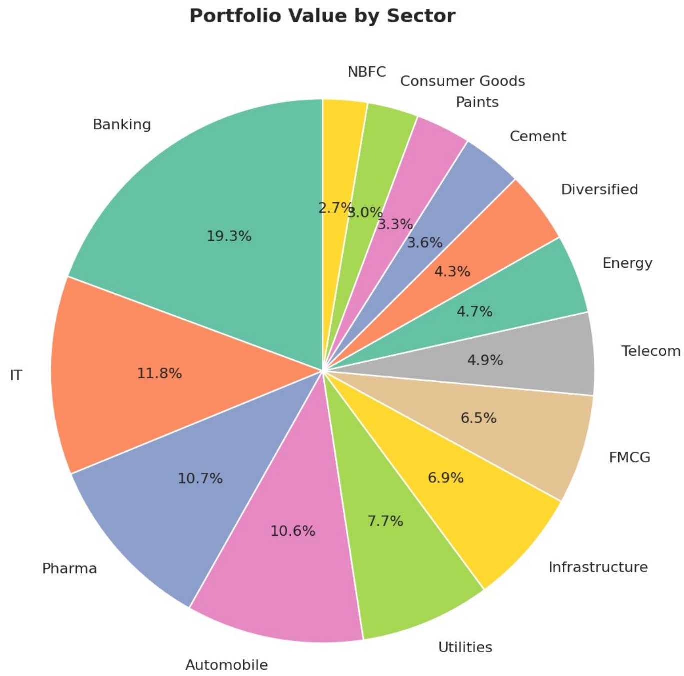


Figure 6 — Aggregate portfolio value by sector across all equity holdings

4.7 Investor Demographics

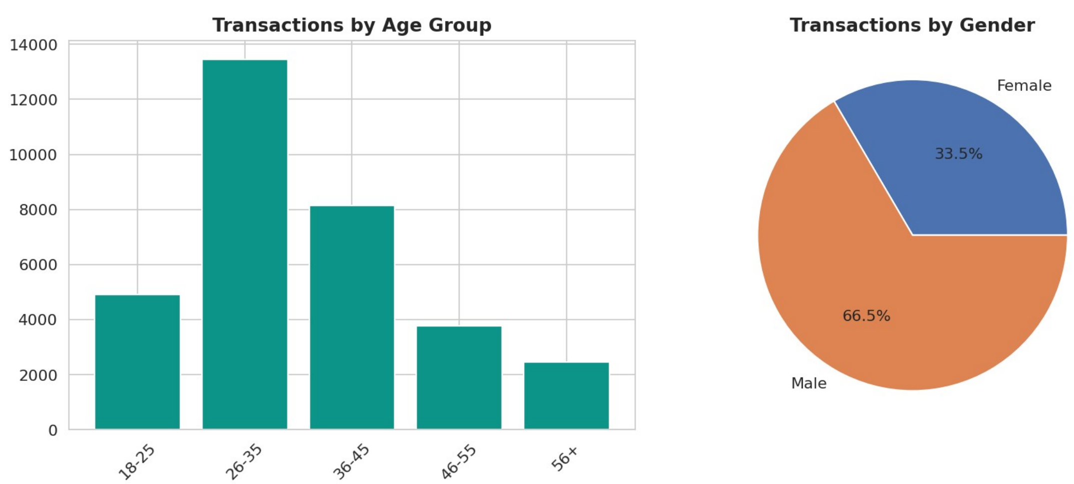


Figure 7 — Transactions by age group and gender

The 26-35 age group drives the highest transaction volume of any cohort, and male investors account for approximately 66.5% of transactions — a demographic skew that may represent an opportunity for targeted financial-inclusion outreach toward underrepresented segments.

4.8 Additional Findings

- SBI Mutual Fund leads the industry in AUM, followed by ICICI Prudential and HDFC; the top three fund houses hold a disproportionate share of industry assets.

- SIP transactions are the most frequent transaction type by count, but Lumpsum investments contribute the largest total transaction value — investors use SIP for steady contributions and lumpsum for larger, one-off allocations.
- Debt fund expense ratios cluster around 0.6-0.8%, roughly half of Equity funds (1.4-1.6%), consistent with lower active management costs for debt strategies.
- KYC compliance stands at 92% of transactions from verified investors, leaving an 8% pending segment representing a potential compliance and onboarding gap.

5. Performance & Risk Analysis

Beyond standard EDA, a dedicated risk analytics pass was conducted to quantify downside risk, track risk-adjusted performance over time, understand investor cohort behavior, and flag portfolio concentration risk.

5.1 Value at Risk & Conditional Value at Risk

Historical VaR (95%) was computed as the 5th percentile of each fund's daily return distribution; CVaR (Expected Shortfall) as the mean return conditional on breaching that threshold.

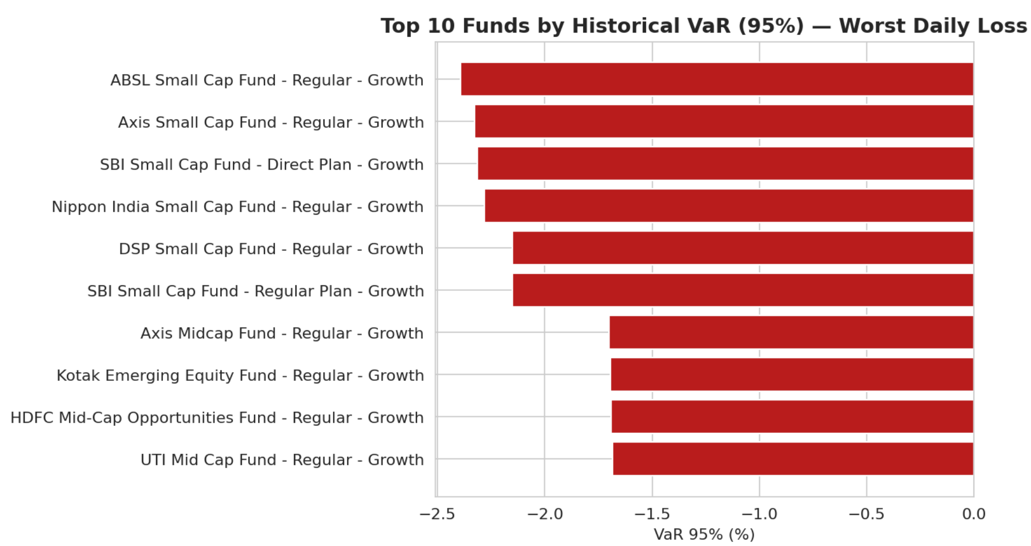


Figure 8 — Top 10 funds by historical VaR (95%), worst expected daily loss

Small-cap equity funds dominate the highest-VaR list, with daily downside at the 95% confidence level reaching roughly -2.3% to -2.4% — nearly double the loss threshold typical of large-cap peers. This is directionally consistent with the risk-return findings from the EDA and confirms small-cap exposure as the primary source of tail risk in this fund universe.

5.2 Rolling 90-Day Sharpe Ratio

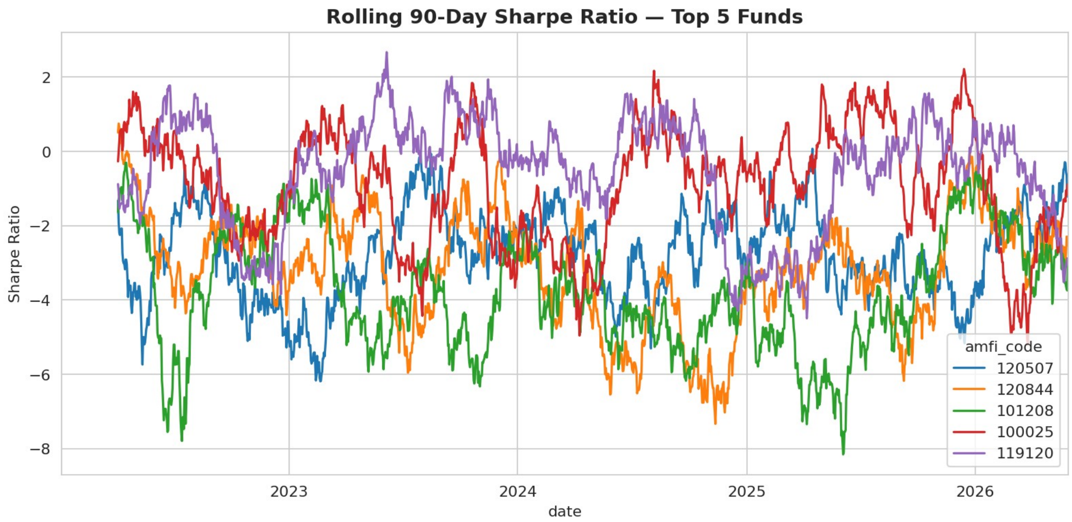


Figure 9 — Rolling 90-day Sharpe ratio, top 5 funds by trailing Sharpe

Rolling Sharpe ratios for the top 5 funds swing widely across the observed window — ranging roughly from -6 to +3 — indicating that no single fund maintained consistently strong risk-adjusted performance over time. This is an important caveat for fund selection: a fund's trailing (point-in-time) Sharpe ratio can be a poor predictor of its near-term risk-adjusted performance, and selection decisions should weigh a longer track record rather than a single snapshot.

5.3 Sector Concentration (HHI)

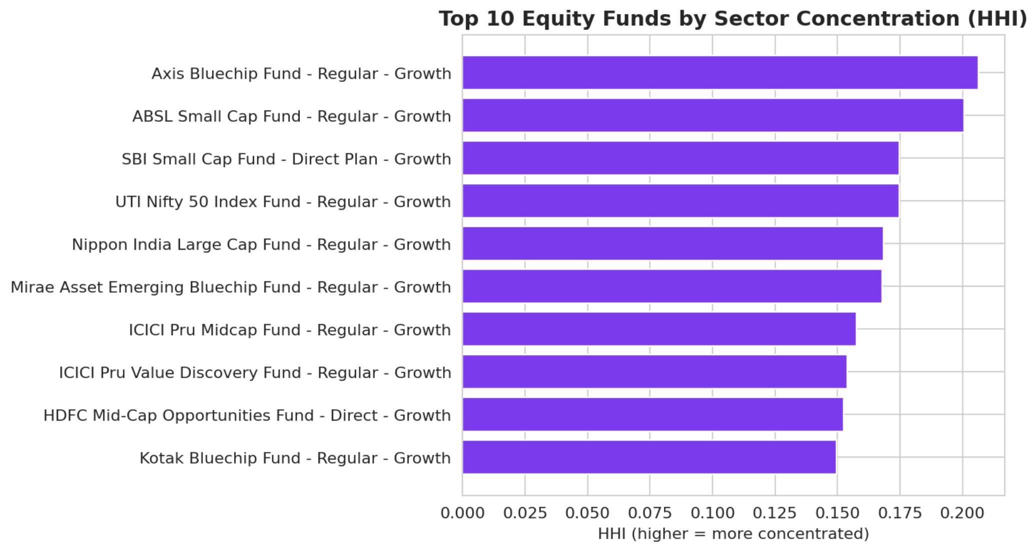


Figure 10 — Top 10 equity funds by Herfindahl-Hirschman Index (sector concentration)

The Herfindahl-Hirschman Index (sum of squared sector weights) was computed for every equity fund's portfolio. Axis Bluechip Fund shows the highest concentration (HHI ≈ 0.206), suggesting meaningfully higher single-sector exposure than its 'diversified large-cap' category label might imply. This finding is useful for investors who assume category labels alone guarantee diversification.

5.4 Investor Cohort Analysis

Cohort (First Txn Year)	Total Invested (₹)	Avg SIP Amount (₹)
2024	₹349.11 Cr	₹10,996.89
2025	₹3.05 Cr	₹13,505.21

The 2025 cohort invests more per SIP transaction on average (₹13,505 vs ₹10,997 for 2024), but the 2024 cohort accounts for the overwhelming majority of total capital deployed to date (₹349 Cr vs ₹3 Cr). This reflects that most of the current investor base joined and built up their holdings in 2024, while 2025 entrants are still in an earlier accumulation phase, albeit with somewhat larger individual contribution sizes.

5.5 SIP Continuity

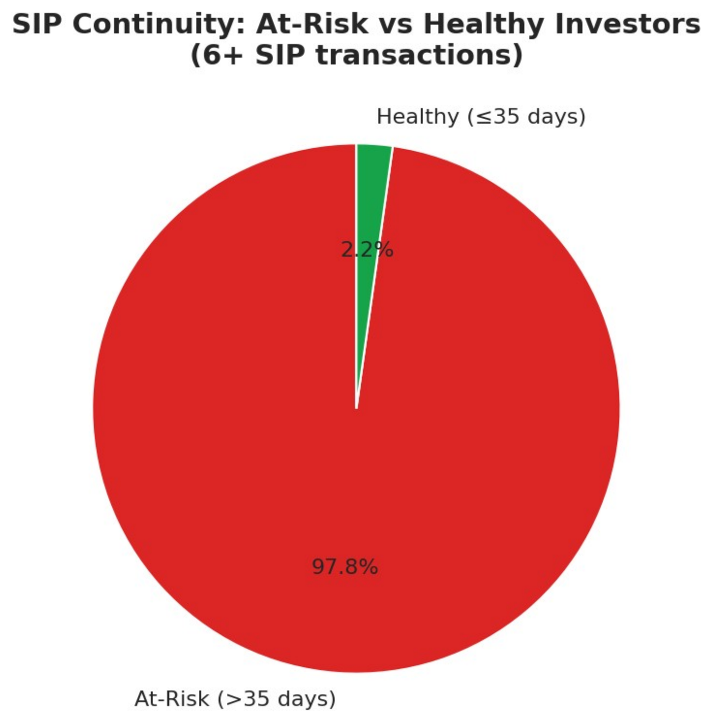


Figure 11 — SIP continuity among investors with 6+ SIP transactions

Among investors with 6 or more SIP transactions, the median gap between contributions is 64.7 days — nearly double the 35-day monthly-SIP benchmark used to flag risk — and 97.8% (1,332 of 1,362) are flagged as at-risk under this threshold. This is a systemic pattern rather than a small edge case, and merits deeper product investigation: it may reflect genuine SIP attrition or pausing behavior, or a broader-than-monthly intended contribution schedule across the investor base that the current 35-day benchmark does not account for.

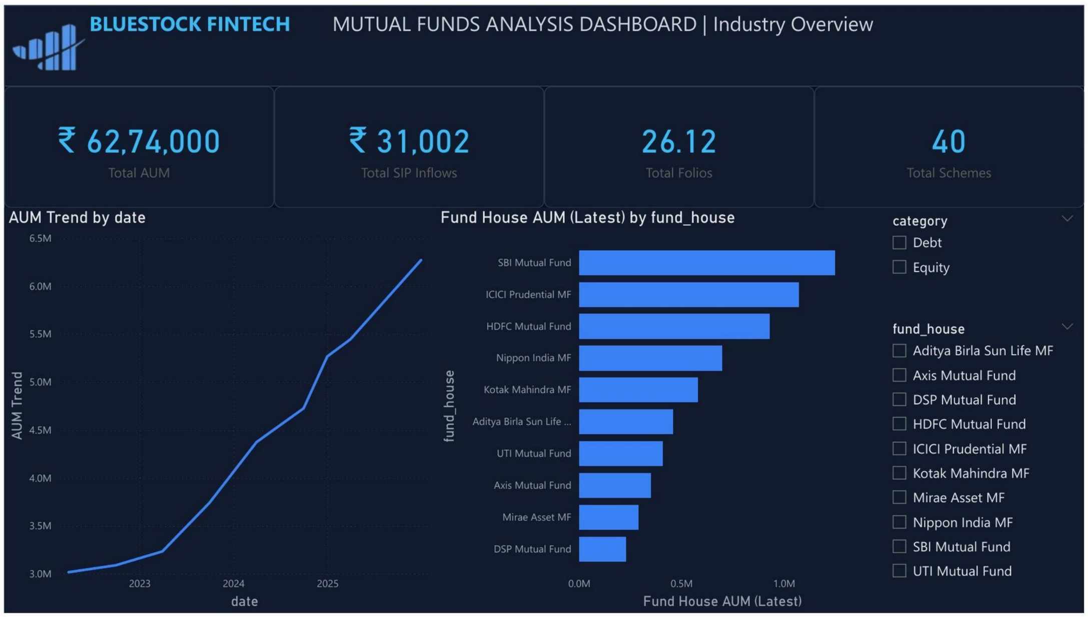
5.6 Fund Recommender

A rule-based fund recommendation engine (recommender.py) was built on top of the fact_performance table. Given an investor's stated risk appetite (Low, Moderate, or High), it returns the top 3 funds by Sharpe ratio within the matching risk_grade category. This provides a simple, explainable first pass at personalized fund suggestions and can be extended with additional filtering criteria (expense ratio ceilings, minimum AUM, exclusion of specific fund houses) in future iterations.

6. Dashboard

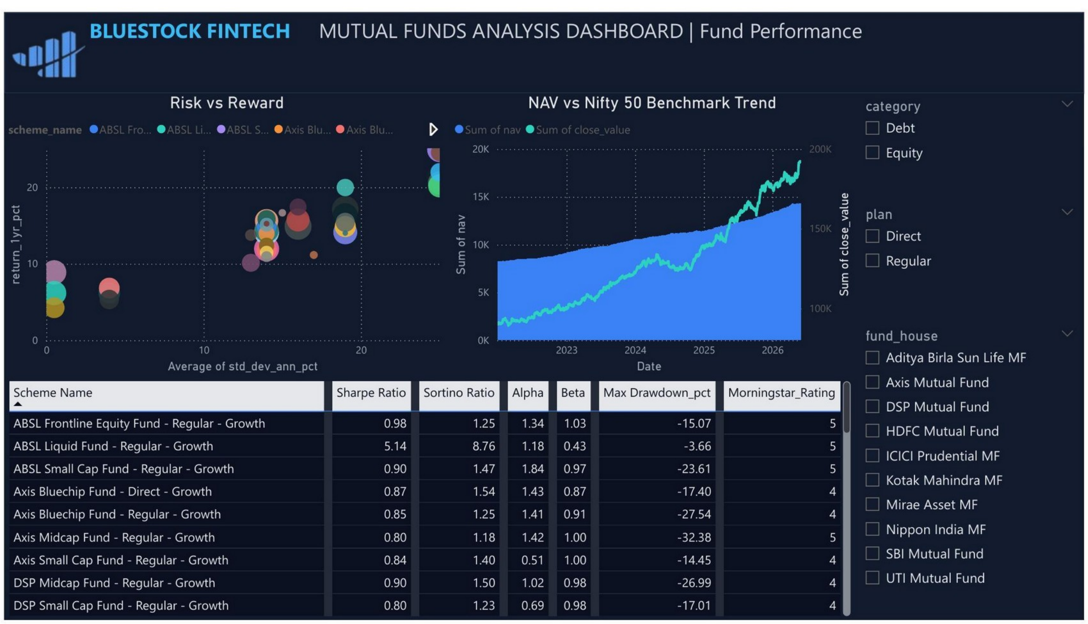
An interactive four-page Power BI dashboard was built on top of the same data model, giving stakeholders a self-service view into industry trends, fund performance, investor behavior, and SIP/category flows. All four pages support cross-filtering via category, fund house, plan type, state, and age group slicers.

6.1 Industry Overview



Total AUM ₹62.74 lakh Cr, ₹31,002 Cr SIP inflows, 26.12 Cr folios, 40 schemes tracked. SBI Mutual Fund leads fund-house AUM, consistent with the EDA finding in Section 4.8.

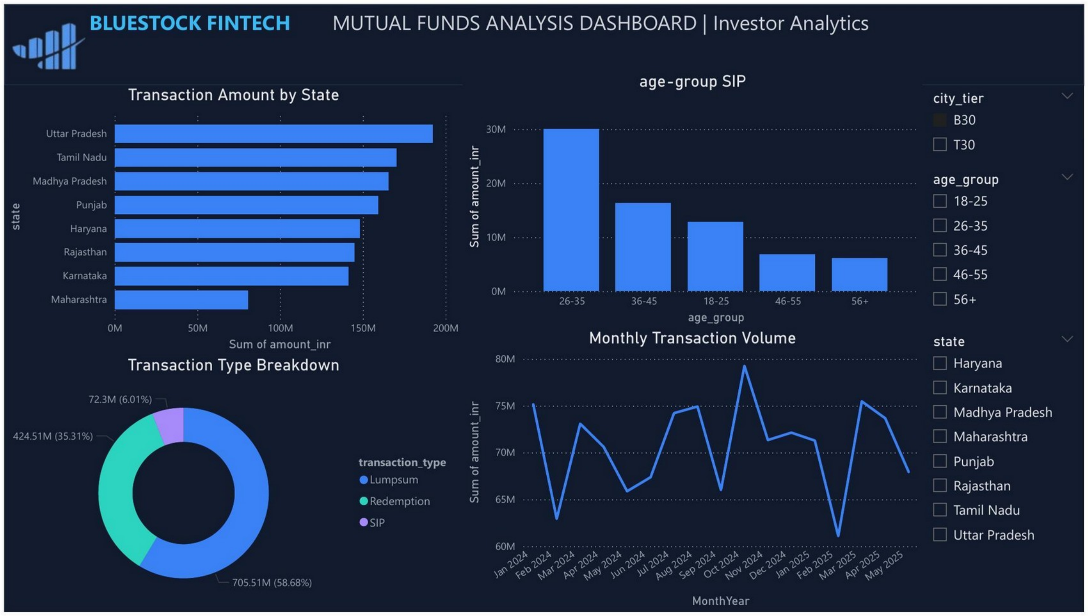
6.2 Fund Performance



Scheme Name	Sharpe Ratio	Sortino Ratio	Alpha	Beta	Max Drawdown_pct	Morningstar_Rating
ABSL Frontline Equity Fund - Regular - Growth	0.98	1.25	1.34	1.03	-15.07	5
ABSL Liquid Fund - Regular - Growth	5.14	8.76	1.18	0.43	-3.66	5
ABSL Small Cap Fund - Regular - Growth	0.90	1.47	1.84	0.97	-23.61	5
Axis Bluechip Fund - Direct - Growth	0.87	1.54	1.43	0.87	-17.40	4
Axis Bluechip Fund - Regular - Growth	0.85	1.25	1.41	0.91	-27.54	4
Axis Midcap Fund - Regular - Growth	0.80	1.18	1.42	1.00	-32.38	5
Axis Small Cap Fund - Regular - Growth	0.84	1.40	0.51	1.00	-14.45	4
DSP Midcap Fund - Regular - Growth	0.90	1.50	1.02	0.98	-26.99	4
DSP Small Cap Fund - Regular - Growth	0.80	1.23	0.69	0.98	-17.01	4

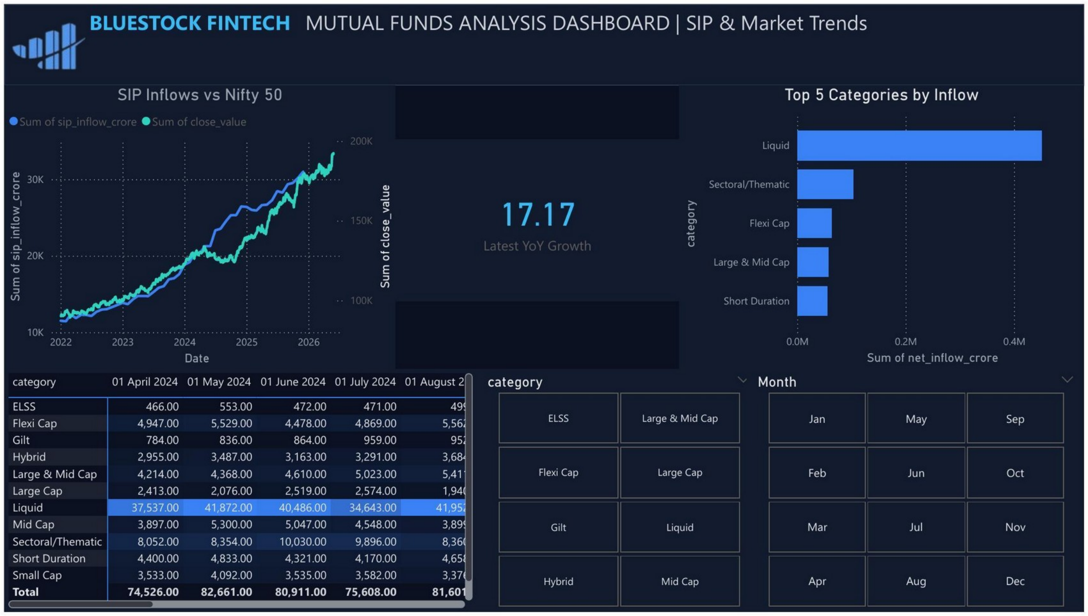
Risk-vs-reward scatter, NAV vs Nifty 50 benchmark trend, and a sortable performance table with Sharpe, Sortino, Alpha, Beta, Max Drawdown, and Morningstar Rating per scheme.

6.3 Investor Analytics



Uttar Pradesh leads transaction value by state; Lumpsum investments account for 58.7% of transaction value vs 35.3% Redemption and 6% SIP; the 26-35 age group drives the highest SIP volume.

6.4 SIP & Market Trends



SIP inflows tracked against the Nifty 50 benchmark; Liquid, Sectoral/Thematic, and Flexi Cap lead category-wise net inflows; category-by-month inflow breakdown available via the detail table.

7. Recommendations

- **Investigate SIP attrition drivers.** With 97.8% of regular SIP investors showing gaps exceeding 35 days, a targeted retention study — segmented by age group, city tier, and payment mode — should identify whether this reflects genuine lapses, seasonal skipping, or simply a longer intended contribution cadence than assumed.
- **Flag concentrated funds more clearly to investors.** Funds like Axis Bluechip with meaningfully above-average HHI should carry clearer diversification disclosures, since category labels alone (e.g. 'Large Cap') do not guarantee low sector concentration.
- **Caution against trailing-Sharpe-only fund selection.** Given the volatility observed in rolling 90-day Sharpe ratios, marketing and recommendation logic should weight longer track records over single-point Sharpe snapshots, and the recommender engine should be extended to incorporate a trailing window rather than a static value.
- **Expand the KYC-pending investor segment.** The 8% of transactions from investors without completed KYC represents a straightforward onboarding-completion opportunity with direct compliance and AUM-growth upside.
- **Target underrepresented demographics.** With male investors at 66.5% of transaction volume and the 26-35 age group dominant, financial literacy and outreach campaigns aimed at women investors and older age brackets (46+) represent an underpenetrated growth segment.
- **Extend the recommender with additional filters.** Incorporating expense ratio ceilings, minimum AUM thresholds, and exclusion lists would make the current Sharpe-only recommender materially more production-ready.

8. Limitations

- Sample size and time span: the underlying dataset spans a relatively short observed window (cohorts only present for 2024-2025), which limits the reliability of longer-horizon cohort and retention trend conclusions.
- SIP continuity threshold: the 35-day at-risk threshold is a reasonable but somewhat arbitrary benchmark for 'monthly' SIP behavior; the true intended contribution cadence per investor is not directly known from the available data, so the 97.8% at-risk figure should be read as directional rather than a precise attrition rate.
- Static risk-free rate assumption: Sharpe ratio calculations assume a flat 6% annual risk-free rate rather than a period-accurate benchmark (e.g., contemporaneous T-bill yields), which may modestly shift absolute Sharpe values, though relative fund rankings are expected to be robust to this.
- Recommender simplicity: the current fund recommender ranks purely on Sharpe ratio within a risk grade; it does not yet account for expense ratio, liquidity, minimum investment thresholds, or an investor's existing portfolio composition.
- Synthetic/representative data caveats: as with many capstone-scale projects, some transaction and demographic fields may not perfectly reflect real-world investor population distributions, and conclusions drawn (e.g., demographic skew) should be validated against production data before informing real business decisions.
- Dashboard scope: the current Power BI dashboard is a desktop (.pbix) file; it has not yet been published to Power BI Service or an equivalent shareable web platform, which is called out as an optional next step.

Despite these limitations, the project delivers a complete, reproducible pipeline — from raw data through cleaning, modeling, analysis, and visualization — that surfaces genuine, actionable patterns in fund performance and investor behavior, and provides a solid foundation for further iteration.